1. Consider the following proposition, which we could label as $P(n)$.
   If $5n + 3$ is even, then n is odd.

   Prove by strong induction that for every positive integer n, proposition $P(n)$ is true.

   Note that I don't think this holds as k might be getting redefined
   Proof :
   By strong induction, assume that $P(k)$ is true for $1 \leqslant k \leqslant n - 1$
   The base case of $n = 2$ ($n = 1$ produces an odd value for $P(n)$, so it is not a base case) holds as $P(2) = 5(2) + 3 = 13$ which is odd, and 2 is even.
   By contrapositive, assume $k + 1 = 2b$ where b is an integer
   $k = 2b - 1$, which is a contradicition thus proving the inductive step through contrapositive. ■

2. Prove by strong induction that for every positive integer n the following holds:

$\sum\limits_{k=1}^{n} k(k+1) = \frac{n(n+1)(n+2)}{3}$

proof:

By strong induction, assume:

$\sum\limits_{k=1}^{n-1} k(k+1) = \frac{n(n+1)(n-1)}{3}$

$\sum\limits_{k=1}^{n} k(k+1) - (n-1)((n-1)+1) = \frac{n(n+1)(n+2)}{3}$ By subsituiting the last term of the summation

$$\left(\frac{n(n+1)(n+2)}{3}\right) - (n-1)((n-1)+1) = \frac{n(n+1)(n+2)}{3}$$

$n(n+1)(n+2) - 3(n-1)((n-1)+1) = n(n+1)(n+2)$

$n^3 + 3n^2 + 2n - (3n+3)((3n+3)+-3) = n^3 + 3n^2 + 2n$

$n^3 + 3n^2 + 2n - 3n^2 - 3n = n^3 + 3n^2 + 2n$

$n^3 - n = 3n^2 + 2n$

There's a mistake in my algebra I can't place, but I'm confident that this is the right approach.